

RecycleBot: Efficient Waste Detection and Classification for Sustainable Recycling



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ABSTRACT

Towards the goal of improving sustainability and developing practical solutions in tackling the environmental challenges associated with reducing the carbon footprint of our everyday actions, the project is aimed at improving the recyclable content of the household trash. Even with good intentions of recycling, one often faces a conundrum whether a certain item is recyclable or not. In real-world situations, waste disposal can be complex, with different types of waste items mixed in a cluttered manner and often in deformed states. To assist in accurate classification of a trash item into recycle or organic category, two-step AI-driven approach, integrating Vision Transformers (ViTs) for image segmentation and Convolutional Neural Networks (CNNs) for classification

Image segmentation model was trained on a 4500 waste images with different levels of diversity and complexity. Transfer learning based on VGG16 image classification model enhanced with three deep layers was trained on 22,500 images of organic and recyclable objects. The model with training accuracy of 93% was applied on a test dataset of 2511 images to achieve F1 score (measure of false positives and false negatives) of 83%. An optimal learning rate of 0.01 ensured stability and improved accuracy. This model is implemented as a web application RecycleBot, which was used on several field images to demonstrate the benefits of using CNN based classifier to improve the fraction of the recycling objects correctly, supporting the hypothesis. The capabilities developed in this project show great potential in automating waste management.

INTRODUCTION

Background

The rapid increase in disposable goods across various industries has led to a significant rise in municipal solid waste (MSW), presenting numerous environmental challenges. In 2018, the United States generated approximately 292.4 million tons of MSW, averaging about 4.9 pounds per person per day. This waste stream includes everyday items such as packaging materials, food scraps, furniture,

and electronics, encompassing both hazardous and non-hazardous materials. Effective management of MSW is crucial to maintaining ecological balance, which has been increasingly disrupted over the past few decades [6].

Recycling is an important mechanism to improve the sustainability of our ecosystem and despite increased awareness, only 13% of trash is currently recycled on the global level. Globally, 400 million tons of plastic waste is produced annually. In the US only 6% of 40 million tons of plastics produced are recycled. Specifically, 85% of plastic packaging goes to landfill. In the US between 2019 and 2020, plastics recycling decreased by 5.7% [4].

In addition to preventing pollution, recycling helps in preserving natural resources, saves energy, and protects the environment and human health, e.g. it takes 95% less energy to recycle aluminum than it does to make it from raw materials. As a leading example, Sweden only puts 1% of its waste in the landfill and recycles rest into reuse and generating energy. The rest of the world is lagging significantly in recycling metrics [11].

A major obstacle in waste management is the widespread confusion about which items are recyclable. This uncertainty leads to improper disposal, where recyclable materials are often mixed with non-recyclables, complicating the recycling process. Such contamination can result in entire batches of recyclables being diverted to landfills, thereby diminishing the efficiency of recycling efforts. Improper separation not only causes the loss of valuable materials but also places an additional burden on sustainability initiatives. When waste is excessively mixed, the recycling process becomes less efficient and more labor-intensive, diluting the recovery of precious materials.

Furthermore, the overaccumulation of waste in landfills poses significant environmental hazards. Hazardous chemicals can leach into groundwater, and the decomposition of organic materials produces methane—a potent greenhouse gas contributing to climate change. Implementing comprehensive waste diversion practices can significantly reduce greenhouse gas emissions. For instance, adopting effective waste management and recycling solutions worldwide could lower

emissions by between 2.1 and 2.8 billion tons of CO₂ per year by 2030, accounting for around 5% of global GHG emissions.

Addressing these challenges requires a multifaceted approach. Educating and empowering consumers on proper recycling practices and tools can reduce confusion and improve waste separation at the source. Implementing advanced sorting technologies can efficiently separate mixed waste, recovering valuable materials and reducing contamination. By adopting these strategies, we can enhance the efficiency of recycling programs, conserve resources, and mitigate the environmental impacts associated with improper waste management. The challenge is not new; however, the opportunity is to leverage emerging technologies to address this [11].

AI-Enabled Waste Classification

Artificial Intelligence (AI) is revolutionizing waste management by enhancing efficiency, accuracy, and sustainability [4,11]. Traditional waste sorting methods often face challenges such as contamination and inefficiency, leading to increased landfill usage and environmental harm. AI addresses these issues by automating and optimizing waste processing. AI-powered systems can accurately identify and separate various waste materials, reducing contamination and increasing recycling rates. This automation not only improves operational efficiency but also minimizes human error and exposure to hazardous materials. Additionally, AI can predict waste generation patterns, enabling more efficient scheduling and routing of collection vehicles, thereby lowering operational costs and environmental impact.[12]

In recent years, machine learning and deep learning techniques have emerged as powerful tools for image recognition. These AI techniques are trained on large amounts of data to identify structures and patterns in images, resulting in improved classification accuracy. In real-world situations, waste disposal can be complex, with different types of waste items mixed in a cluttered manner and often in deformed states. In these situations, an efficient approach involves image segmentation followed by image classification.

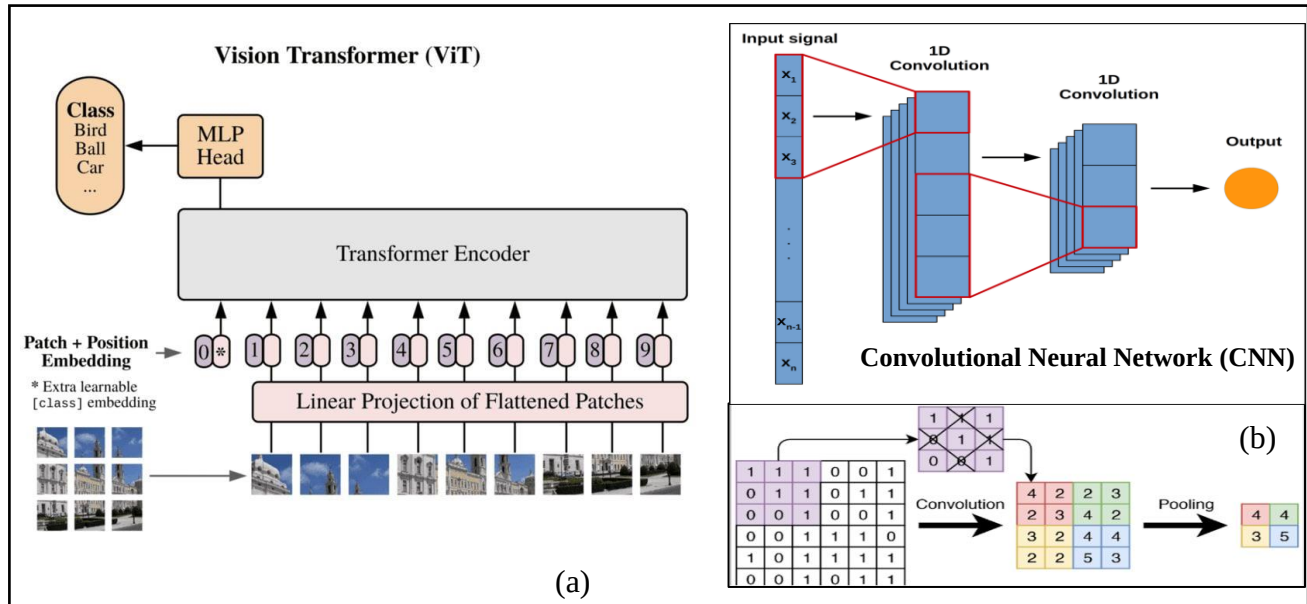


Figure 1: (a) Visual Transformer architecture for image segmentation, (b) Deep learning architectures for waste classification using Convolutional Neural Network (CNN)

Image Segmentation with Vision Transformers (ViT):

For image segmentation, Vision Transformers (ViT) offer a novel approach by dividing an image into a grid of equal-sized patches, like cutting a picture into puzzle pieces. Each patch is then transformed into a numerical representation, capturing its essential information. The ViT processes these patches simultaneously, assessing the relationships and patterns between them to comprehend local details and the broader context of an image, enabling the recognition of complex patterns and structures. (Fig. 1 (a)) This process of semantic segmentation, classifying each pixel or region of image to a particular waste category, is essential for robust waste classification.

Waste Classification with Convolutional Neural Networks (CNN):

For waste classification, a computer vision approach called Convolutional Neural Network (CNN) is applied. A CNN is a type of artificial intelligence designed to process visual information, much like the human brain interprets images. By training these networks on extensive datasets, AI systems learn to recognize and categorize materials such as plastics, metals, glass, and paper. CNN models contain many layers of neurons (combinations of variables and image features) used for building correlations. Each of the convolution layers acts as a specific filter applied to generate a feature map, extracting distinct characteristics or qualities of the original image. (Fig. 1 (b))

Transfer Learning with VGG16:

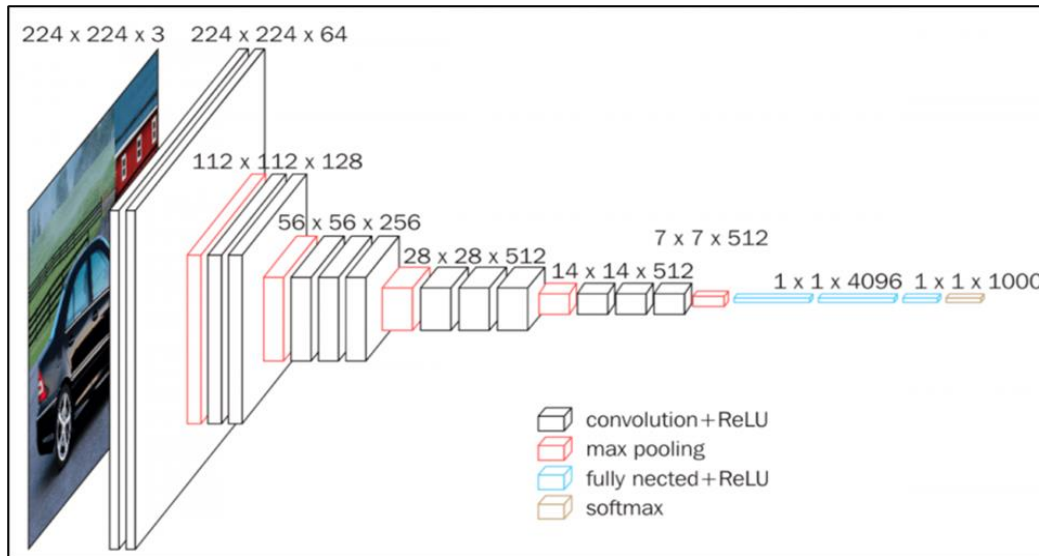


Figure 2: Transfer Learning based on VGG16 for Image Classification

CNN models rely on large amounts of training data to achieve high classification accuracies. However, datasets of this scale are not always available for waste classification. In such situations, transfer learning techniques have been employed to achieve higher prediction accuracy. Transfer learning is a class of deep learning technique in which a CNN model trained on a different dataset is used as a starting model. For instance, the VGG16 algorithm, trained on the ImageNet dataset containing images of fixed size of 224x224 pixels with RGB channels, has been explored for this purpose. Studies have shown that models like VGG16 can achieve high accuracy in classifying various types of trash objects (Fig. 2).

By integrating these advanced AI techniques, waste management systems can become more efficient, accurate, and sustainable, significantly contributing to environmental conservation efforts.

Research Study

The primary focus of this research study is to evaluate the effectiveness of the machine learning models in identifying and classifying trash items under real-world complex situations and optimize the accuracy of the classification by adjusting training parameters. Machine Learning based image segmentation and classification application is designed to identify if something is recyclable or organic. This is achieved in two key steps:

1. Develop a visual transformer based semantic segmentation approach to partition an image into meaningful segments to isolate individual waste items.
2. Identify and implement an efficient algorithm using Convolution Neural Network (CNN) approach to detect and classify sub-images into two categories (organic or recyclable) along with estimating the recycled content of the waste.

For each of the above steps, partitioned test dataset is used for fine tuning the accuracy and speed of the segmentation and classification models by changing the training parameters.

As part of building an accurate and robust system, parametric studies on relevant training parameters were performed to arrive at the best models in terms of minimizing false positives and true negatives.

For the classification step, four distinct parameters were considered in this project, namely, (1) Learning Rate: which indicates the speed at which CNN algorithm moves towards solution (2) Drop-out: % of neurons left out of the prediction (3) Epoch: Number of times the CNN is trained on the full dataset of images (4) Optimization Method: Adam and Stochastic Gradient Descent (SGD) as the search approaches.

This research introduces a unique two-step AI-driven approach for waste classification, integrating Vision Transformers (ViTs) for image segmentation and Convolutional Neural Networks (CNNs) for classification has been proposed. By leveraging the strengths of both architectures, the proposed method aims to enhance the accuracy and robustness of automated waste detection in real-world settings.

Key research questions addressed in this work are (1) How effective is the two-step ViT + CNN approach in detecting & classifying trash items? (2) What impact do algorithm parameters (e.g., learning rate, dropout fraction) and training cycles have on segmentation and classification accuracy?

Towards the goal of developing an AI-enabled waste classification system, current study hypothesizes: If the training data set is consistent, balanced and has sufficient diversity of features representing the different classes and varied levels of complexities, then combining ViT-based segmentation with CNN-based classification in terms of organic and recycle classes enhances overall accuracy by ensuring that only refined, segmented waste objects are classified, reducing misclassifications due to background clutter. The best-in-class model will be implemented in an interactive web app that can be used by everyone at the point of disposal.

Design Space

For the research study, the following two independent variables were used to capture the real-world conditions of the waste stream:

- **Environment Complexity:** This is a categorical variable with two levels representing the cluttered-ness and state of deformation of waste items. Depending on the extent of clutter, obscurity of items and deformation, the waste scenes are labeled as [Low, High].
- **Waste Diversity:** This is a categorical variable based on the diversity in the waste stream [glass, plastic, cardboard, paper, organic, electronic, medical]. Low level represents one or two types of waste items; and High level indicates three or more types.

In designing robust and efficient models for segmentation and classification, several critical parameters are evaluated as part of the independent variables design space. Some of the key model parameters and their ranges or settings are described in table 1.

The performance of an algorithm for a given set of testing/validations images is summarized in a confusion matrix (fig. 3) that displays the number of accurate and inaccurate labeling of waste items based on model predictions [2]. Using the data from confusions matrix, model performance is calculated for the following four **Dependent Variables**:

1. **Accuracy** – measures the overall prediction performance of the model. It is the ratio of total correct instances to the total instances:

$$A = \frac{TP + TN}{TP + FP + TN + FN}$$

2. **Precision**– measures the accuracy of a model's positive predictions (specific to target). It is the ratio of true positive predictions to the total number of positive predictions made by the model:

$$P = \frac{TP}{TP + FP}$$

		PREDICTED	
		Recycle	Organic
ACTUAL	Recycle	TRUE POSITIVE TP	FALSE NEGATIVE FN
	Organic	FALSE POSITIVE FP	TRUE NEGATIVE TN

Figure 3: Confusion Matrix for binary classification of waste items

	Model Parameter	Range	Description
1	Learning Rate [LR]	(0.001, 0.1)	Rate at which the CNN moves towards solution
2	Drop Out [DO]	(0.1, 0.4)	% of neurons left out of the prediction
3	Epoch	(2, 10)	Iterations of model training on full set of images
4	Opt. Method [OPT]	(Adam, SGD)	Search Algorithm used to maximize accuracy

Table 1: Algorithm Options Evaluated for Detection and Classification

3. **Recall** – measures the effectiveness of a classification model in identifying all relevant instances (targets) from a dataset. It is the ratio of the number of true positive (TP) instances to the sum of true positive and false negative (FN) instances:

$$R = \frac{TP}{TP + FN}$$

4. **F1-score** – is used to evaluate the overall performance of a classification model. It is the harmonic mean of precision and recall:

$$F1 = \frac{2 * P * R}{(P + R)}$$

The goal is to develop models that have high model accuracy but at the same time does not give too many false positives or false negatives. This is critical for building a robust prediction and F1 score is the most important metric used in selecting the best model. Additionally, it is important for the model training and predictions to be computationally efficient. Training times are kept as a secondary dependent variable.

Controlled Variables: Models are trained on the same partitions of the data sets, following exactly similar standards for images with resolution of 224x224 pixels, augmentation and preprocessing of images.

Experimental Control: Model accuracy and precision achieved on the training partition is cross validated on the testing partition.

MATERIALS AND METHODS

Data Set

The dataset used in this study for classification was publicly available on Kaggle [8], 22,500 images of organic and recyclable objects. Images for typical recyclable objects are shown in figure 4. In terms of target variable, each image is labeled as – “organic” or “recycle”. A chart summarizing the two classes for training and validation is shown in fig. 5. The graph clearly shows the balanced attribute of this dataset with both the classes almost equally represented in both the training and validation partitions. The validation dataset has 2511 images. Additionally, VGG16 model is trained on ImageNet data set with 14,197,122 annotated images.

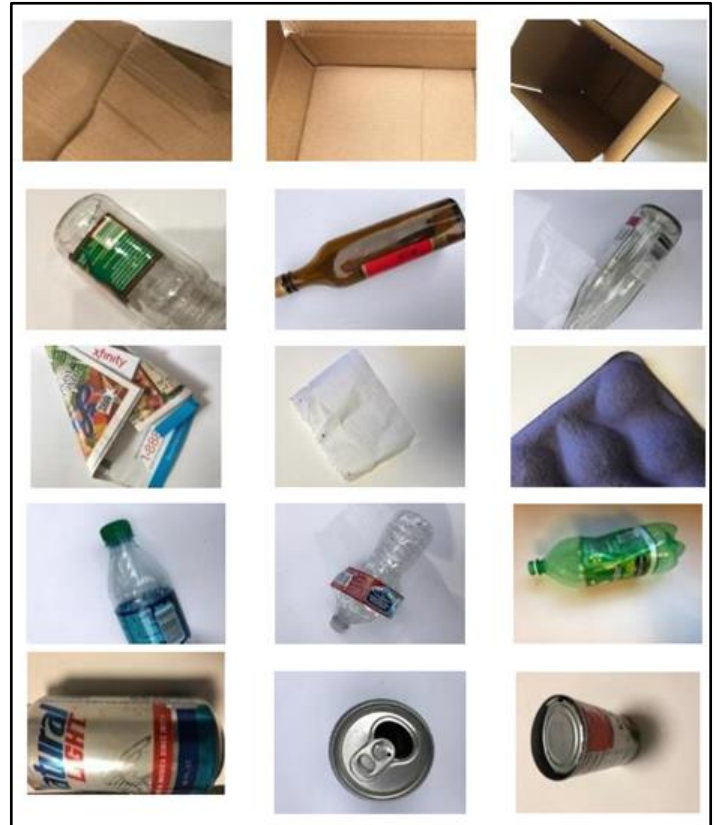


Figure 4: Sample images of recyclable objects (target class)

For training the segmentation model on highly deformable and often translucent objects in cluttered scenes, **Zerowaste** dataset comprising 10,000 high-quality images, including ~4500 images with target labeling [1]. These images depict waste items in complex and cluttered environments, aiming to evaluate the performance of ViT in realistic waste classification scenarios. To experiment and test the performance of different segmentation algorithms in a structured way, all these images were binned in two x-variables: Environment complexity (Low and High) as well as Waste Diversity (Low and High). Within these images, the areas were labeled into 6 classes: Background, Plastic, Metals, Cardboard, paper, organic.

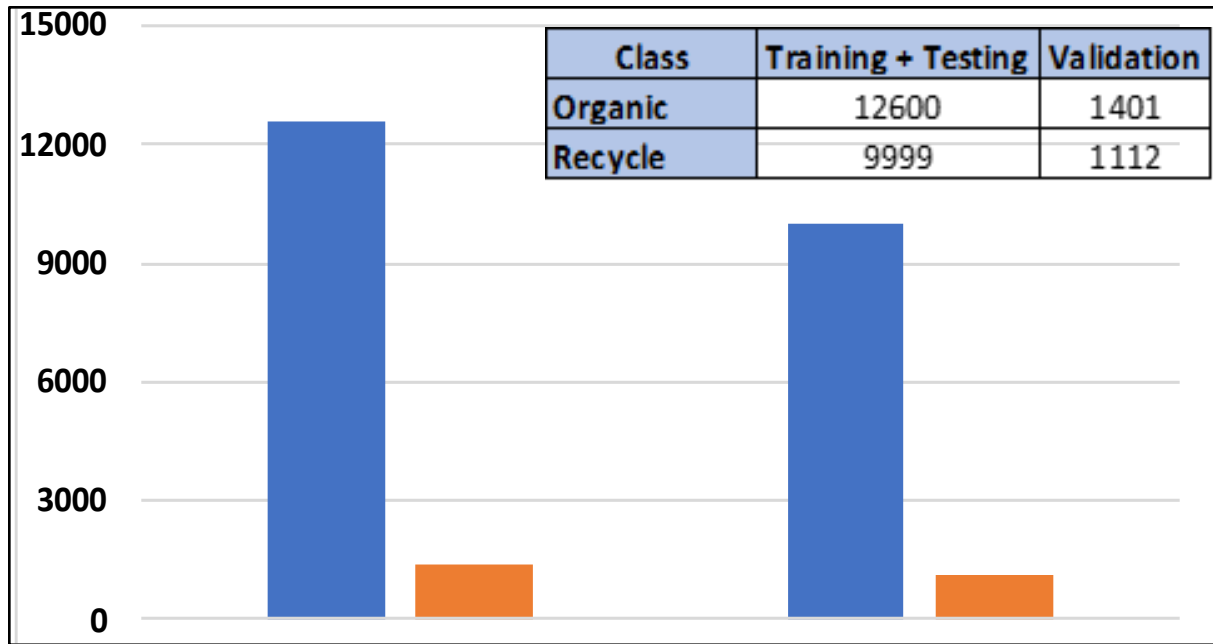


Figure 5: Class summary statistics of Kaggle waste classification dataset

Modeling Environment and Libraries

For model development and testing work, python language was selected for its ease of use and availability of open-source functional libraries [9].

- Modeling and Computing Environment: Google Colab
- Numerical and Statistical Libraries: numpy, pandas, math, sklearn, keras, XgBoost
- Visualization Library: Matplotlib
- Web development environment: Streamlit

A unique 3-stage approach has been developed and implemented to build a robust system for waste classification. The details for each of the three stages, namely, data preparation, model training as well as field validation are described below.

Data Preparation

Data preparation steps are key for building efficient ML/DL models [5]. Model development, execution and testing were performed in Google Colab high RAM python environment. The data sets are

downloaded from Kaggle on Google Drive and files loaded into the model development file using operating system libraries. Following image pre-processing and standardization methods are used.

- **Image Pre-processing and Normalization:** As a first step, images were resized to 224x224 pixels and normalized to scale the intensity values between 0 and 1. For the Transfer learning, all the three channels of the original images were used.
- **Target Labeling:** For the Kaggle dataset, type labels were changed to numerical values by assigning 0 to organic, 1 to recycle. For the zero-waste data set, areas within the images were labeled in six target classes (0-5) for the (background, plastic, metals, cardboard, paper and organic respectively).
- **Data Augmentation:** Size of datasets is critical for improving the predictive power of the models. Towards this cause, horizontal reflection was used to double the size of the dataset.
- **Data Partition:** The dataset was divided into 80% and 20% for training and testing, respectively based on randomized sampling.

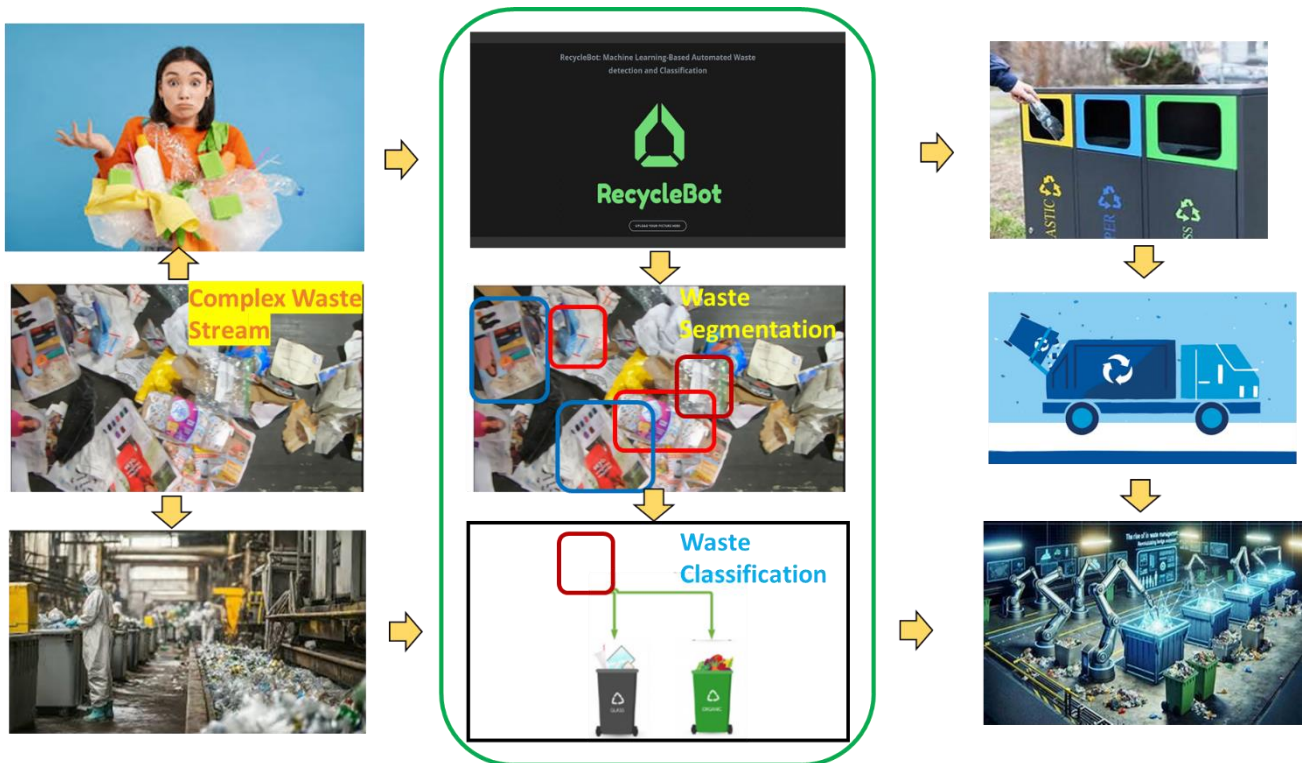


Figure 6: Two-step AI-enabled process for waste segmentation and classification

Model Training and Optimization

The model is developed and tested in python. Figure 6 describes the a two-step AI-enabled process for waste segmentation and classification, leveraging advanced technologies like Vision Transformers (ViTs) for segmentation and Convolutional Neural Networks (CNNs) for classification.

The objective of the first step is to detect and isolate individual waste items from a complex waste stream, separating relevant objects from cluttered backgrounds. A Vision Transformer (ViT)-based segmentation model is applied to analyze the input images from the zerowaste dataset with labeled areas. The model identifies regions of interest (ROIs) corresponding to distinct waste items. Each waste item is highlighted with bounding boxes or masks, separating it from the surrounding clutter. The segmented regions, containing individual waste items, are isolated and prepared for further classification. In the classification step, VGG16 model with 16 layers of neurons is imported from keras library and additional layers are added to build a classifier for the binary target: Recycle vs. Organic. The flatten layer converts the output of CNN into a 1-dimensional array, input to fully connected /dense layer. In the final layer, the sigmoid activation normalizes the output of the network to a probability of an item being recycled. One of the key steps in converging to the optimal training parameters for the model is to perform parametric study to see how the performance metrics change with respect to these parameters.

Model Selection, Prediction and Validation

For image segmentation, pixel accuracy metric, which is the ratio of correctly classified pixels to the total number of pixels to select the best model. Classification model selection and prediction based on comparing accuracy for different settings of hyperparameters. Confusion matrices are generated, which are used to compute accuracy, precision, recall, and F1 score for models with different training parameters. The best model is selected which gives the highest accuracy without compromising the F1

score. This model is used in the web application. The best model for waste classification is used to predict the classes for the training data set (which is not used in the model training or cross-validation).

RESULTS

Hyperparameter Optimization

		Waste Diversity	
		Low	High
Environment Complexity	Low	73.7%	69.1%
	High	69.7%	64.3%

Figure 7: Performance of Segmentation model in terms of pixel accuracy for different waste streams

Vision Transformers (ViTs) have several important parameters that influence their performance, efficiency, and accuracy. In this study SeFormer visual transformer was used for its hierarchical transformer-based feature extraction from waste images with diverse object sizes. Also, it is lightweight and efficient in handling cluttered scenes. Patch size, number of layers and attention head parameters were varied to optimize the accuracy of the segmentation model. Patch Size of 16x16, 18 Layers, 11



Figure 8: Transfer Learning model VGG16 training progress with epochs in terms accuracy (Right) and model loss (left) 16

heads were found to be optimal to give a pixel accuracy of ~70% for all 4 types of waste streams as shown in Fig. 7.

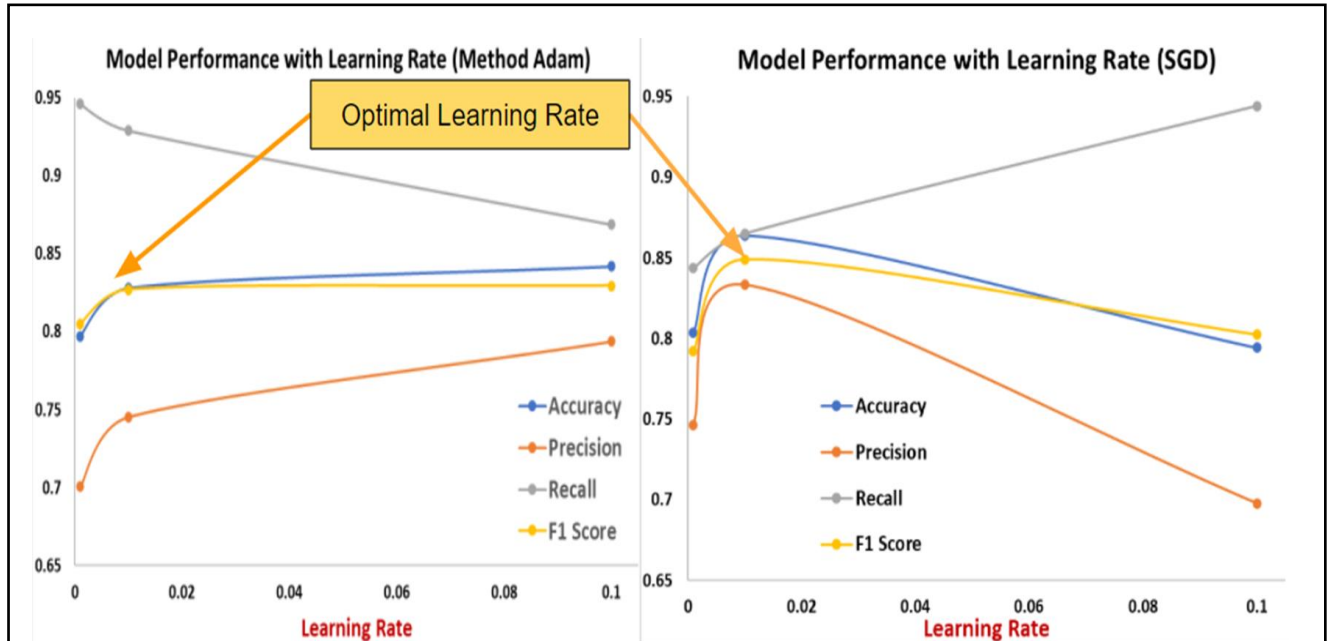


Figure 9: Parametric study w.r.t. learning rate for two different optimizers: Adam (left) and SGD (right)

For transfer learning models, there are several parameters including epochs, dropout and optimizers that were varied to identify the optimal settings. Towards the goal of minimizing the false positives and false negatives, accuracy was used as a key criterion to maximize. Figure 8 shows the variation of accuracy and model loss as a function of the epochs for the VGG16 model for both training and validation datasets.

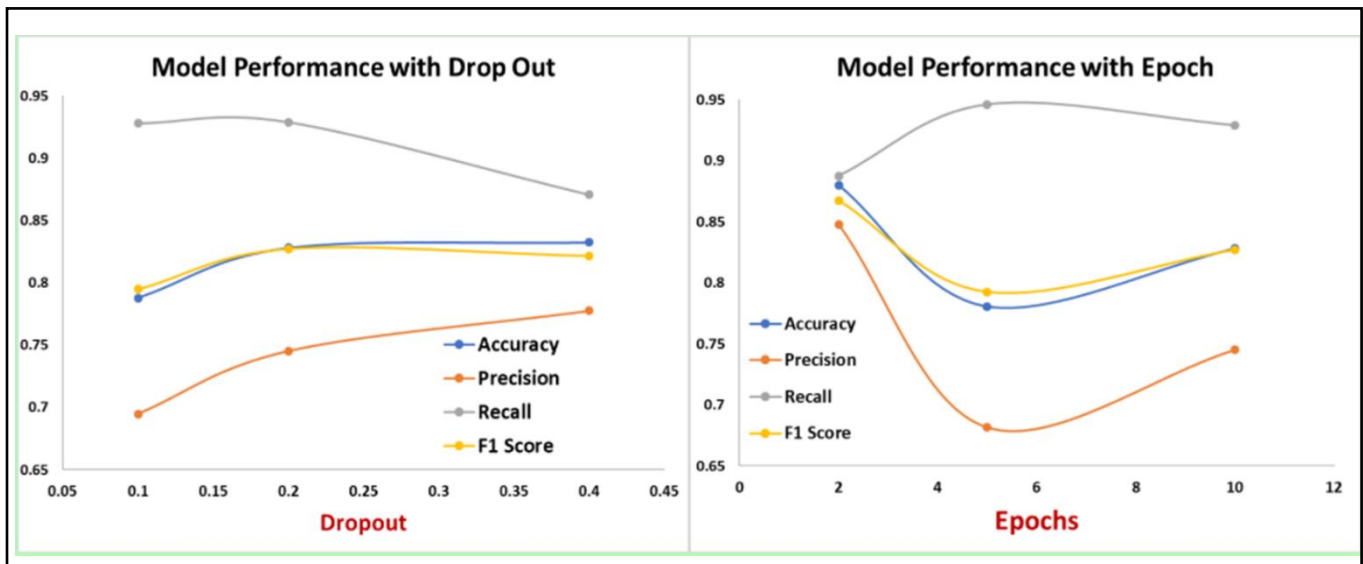


Figure 10: Transfer Learning model VGG16 training performance in terms Dropout (Left) and Epochs (right)

Figure 9 shows the model training with two different optimizer options. Performance metric in terms of F1 score is calculated for different learning rates. Optimal learning is based on maximizing the F1 score for test data. With both algorithms a learning rate of 0.01 gives the best outcome. Figure 10 shows the variation of the performance metrics as a function of the dropout and epochs for the VGG16 model. Five epochs were used as the optimal setting.

For CNN models, several studies were performed with different network architectures to arrive at the optimal setting of the convolution templates, batch size and strides. The best model is selected, and corresponding weights are stored.

Confusion Matrix

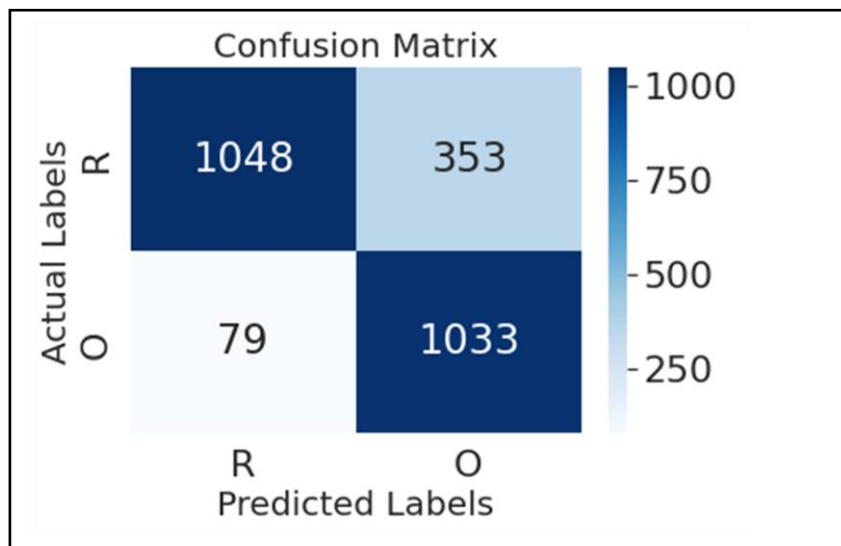


Figure 11: Confusion Matrices for the model predictions on the training dataset.

All the models were trained on standardized data following the same steps for data preparation and using optimal parameters as described in the last section. Trained model was used to predict the class labels for the validation data and a confusion matrix was generated using Scikit Learn. Figure 11 shows the confusion matrices for the most promising AI algorithms for the classification system. The smaller the off-diagonal numbers, the better the F1-score for that model. The most common metric is classification accuracy; However, final classification accuracy is most reliable when the dataset contains an equal number of samples from each class. The unbalanced nature of the dataset requires evaluation of performance metrics.

Performance Metrics

From the confusion matrix, the precision, recall, and F1-score were calculated for each of the algorithms. Table 2 summarizes the performance metrics for the validation data. VGG16 model augmented with three additional layers with optimal algorithm parameters resulted in a very high F1 score of 0.83 on the validation data as well as highest precision of 0.93. The training and testing times per epoch were also recorded and none of them were exceptionally high for the classification problem. Recall is the most critical statistic for improving the recycle content of the trash. Even high precision is important for making sure that organic is not mixed up with the recycled items. High accuracy is not always desirable as that may be based on overfitting, and it leads to lower F1 score. A dropout value of 0.2 is ideal.

Accuracy	$= \frac{TP + TN}{TP + FP + TN + FN} = \frac{1048+1033}{1048+79+353+1033} = \mathbf{82.8\%}$
Precision	$= \frac{TP}{TP + FP} = \frac{1048}{1048+79} = \mathbf{93\%}$
Recall	$= \frac{TP}{TP + FN} = \frac{1048}{1048+353} = \mathbf{74.8\%}$
F1 Score	$= \text{Harmonic Mean } (.93, .748) = \mathbf{82.7\%}$

Table 2: Waste classification metrics for validation data

Pixels		Predicted	
		Recycle	Organic
	Actual		
	Recycle	 	 
	Organic	 	   

Figure 12: Saliency maps for the CNN detection model for no-tumor (left) and tumor (right) cases.

Field Trial

Before deploying the solution, the best model was tested on several field images to demonstrate the benefits of using a classifier to improve the fraction of the recycling objects correctly. Field data samples from household and community trash are collected and the best model was applied for classification. Pictures for each of the items are taken in a consistent setting and they are uploaded in the web app to make a prediction. True class and predicted class are compared in the confusion matrix (Fig. 12). Model also provides the probability value for a given item to be recyclable, which is key for the user confidence. The capabilities developed in this project show great potential in automating waste management.

DISCUSSION AND CONCLUSIONS

Web Application

Best-in-Class models for waste classification were implemented in a novel, easy-to-use web application, RecycleBot. Streamlit library, which allows for easy linking of different python models was used to create a web interface with different end-user options. Figure 13 shows the application interface which is used to predict the class of a trash item in two simple steps:

- Upload a picture of a trash item from the computer or smart phone.
- Model is used to predict class: Recycle or organic and associated probability to be in the predicted class.

Users can choose to check the box on the left to get the model performance metrics, confusion matrix and explainability plots associated with the prediction.

Image segmentation and classification approach using a combination of Visual Transformer (ViT) and Convolutional Neural Network (CNN) models is effective in classifying trash items into organic and recycling classes. Data supported the original hypothesis that the transfer-learning based Convolutional

Neural Network (CNN) classification models can be built on the training data which will effectively classify trash items. Models with larger number of parameters will perform better in terms of the accuracy of the classification error, however it will have a lower specificity because of overfitting, thus requiring a balanced explainable approach for building confidence in predictions. Furthermore, this novel approach provides the recycle content of the real-world cluttered waste stream to help the users of the app to make an optimal decision in recycling as well as efficient waste management collection at the community level.

VGG16 enhanced with three deep layers performs at greater than 90% accuracy in predicting the recycle objects for training data. The classification model was applied on a validation data of 2511 images achieving accuracy of 83% (17% false positive & negatives). Learning rate is an important parameter in model training and an optimum value of 0.01 was used to arrive at the best predictive model (F1 score of 83%). Drop Out of 0.2 is optimal.

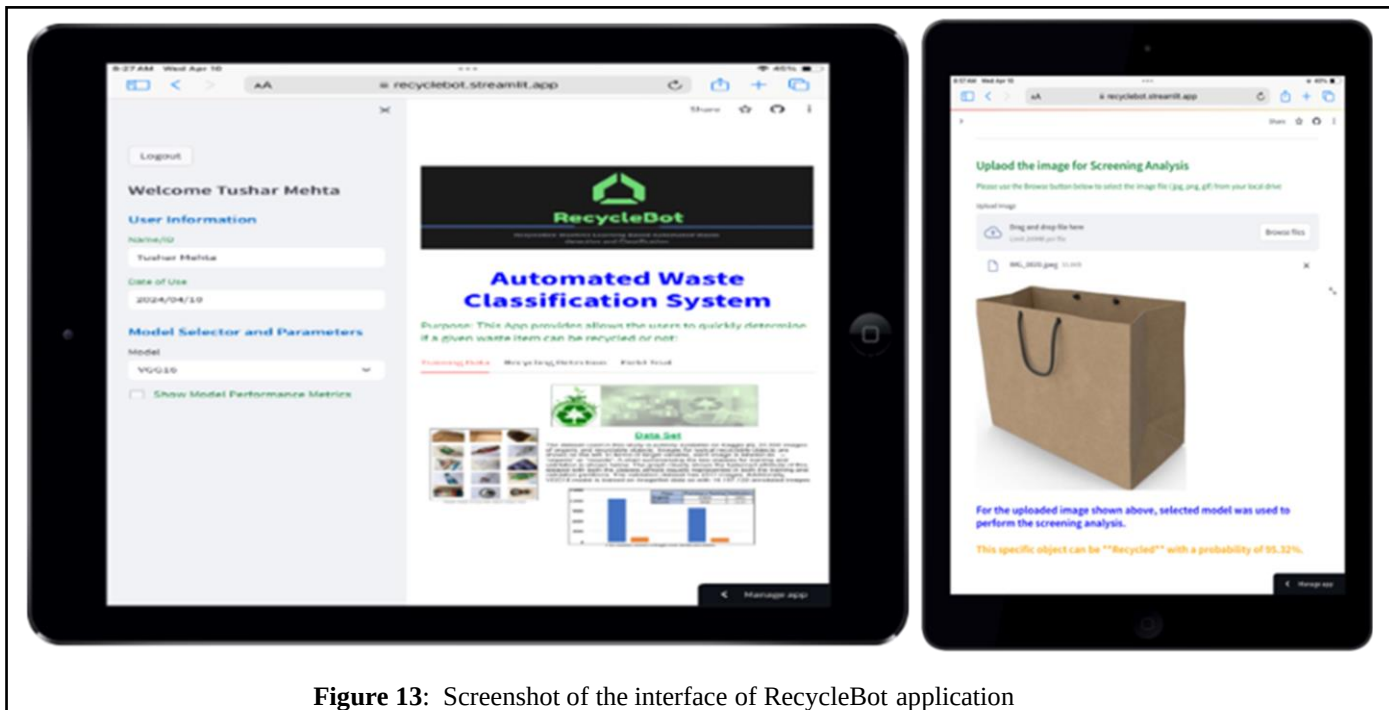


Figure 13: Screenshot of the interface of RecycleBot application

As a final product of this work, best-in-class models for detection and classification were implemented in a novel, easy-to-use waste classification application, **RecycleBot**. It was used on several field

images to demonstrate the benefits of using classifier to improve the fraction of the recycling objects correctly.

Future Research

To improve the model robustness, images from the field trial could be included in training as well as there may be value in including additional data sets. As a next step, F1 score of the classifier can be improved by including multiple views of an object in the training as well as standardizing the image capture process in terms of zoom level, background, and lighting. Another idea for potential improvement in the F1-score is the fusion of diverse features from multiple transfer learning approaches. Besides improving the predictive power of the classifier, classification models can be extended to include other types of trash items to build comprehensive waste management tool [10]. To improve the confidence and explainability of the predictions, there is an opportunity to further extend the feature-extraction based models based on multiple objects in a picture. This can be accomplished by implementing image segmentation in composite pictures as a precursor to image classification. To realize the true potential of this technology, there is work required to integrate RecycleBot into recycle bins to make the waste management activity fully autonomous. Smart recycling bins will enable faster adoption of this approach.

Environmental Applications

- Waste Management: RecycleBot can play a key role in driving ease & fraction of recycle content towards improving sustainability of the environment.
- Biohazard Safety: Model can be customized to identify and segregate medical and infectious waste.
- Recycle Culture: Facilitates awareness & culture of recycling!

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